

DocMind AI

Problem Statement

The Challenge

Organizations, students, researchers, legal teams, educators, and businesses manage large volumes of digital documents including reports, research papers, contracts, manuals, presentations, and notes. Manually reading these documents to find relevant information is time-consuming and inefficient. Traditional search methods often fail to understand context, making document analysis slow and reducing productivity.

The Objective

Develop **DocMind AI**, an AI-powered intelligent document analysis platform that enables users to upload documents and interact with them using natural language. The solution aims to answer document-based questions, generate summaries, compare documents, extract insights, support PDF/DOCX/PPTX/TXT formats, and provide a secure workspace for AI-powered document management.

Proposed Solution

DocMind AI is an AI-powered intelligent document workspace developed using **Python Flask** and **IBM watsonx.ai**. It enables users to upload documents and perform AI-driven analysis through a modern web interface.

The platform uses the **Meta Llama 3.3 70B Instruct** foundation model via IBM watsonx.ai to understand document content and provide context-aware responses. Key capabilities include Document Q&A, AI Summarization, Document Comparison, and Insight Extraction. The application supports PDF, DOCX, PPTX, and TXT files while maintaining conversation history and secure user authentication.

By combining intelligent document processing with IBM's enterprise AI platform, DocMind AI simplifies information retrieval, improves productivity, and enables faster, data-driven decision-making for students, researchers, professionals, and organizations.